

High trait diversity among soil arthropods allows their survival in a heterogeneous eco-agricultural mosaic

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ABSTRACT

Studies examining changes in functional diversity and trait composition for soil arthropods are limited yet crucial for understanding the effects of land-use change. To determine whether plantation forestry drives functional homogenization of soil biota, we compared the taxonomic and functional diversity of ants and springtails between natural (indigenous forests and grasslands) and transformed (*Eucalyptus* plantations) biotopes. The prevalence of morphological and ecological traits in natural vs. transformed biotopes was also assessed. The study was conducted in two environmentally contrasting regions: high elevation areas with clay soils and coastal areas with sandy soils, in KwaZulu-Natal, South Africa. Surprisingly, the *Eucalyptus* compartments generally had similar levels of taxonomic and functional diversity as indigenous forests and/or grasslands. However, there were shifts in traits between the three biotopes, with each biotope supporting arthropods with traits suited to the environmental conditions in the biotope. Some traits were representative of the natural biotopes, and others of the plantation compartments. As a functionally diverse component of soil biodiversity is present in the plantations, it is indicative that these plantations have seemingly high levels of ecosystem stability and functioning. Arthropod traits compatible with the different environments appear to allow them to persist in high diversity across the landscape mosaic. These results suggest that enhancing landscape heterogeneity in a production landscape increases taxonomic and functional diversity of soil arthropods by creating a large-scale, spatial mosaic of complementary conditions for species with different habitat requirements.

1. Introduction

The rapid increase of anthropogenic activities, such as the conversion of natural ecosystems into agricultural systems, can lead to changes in environmental conditions, which can have significant effects on biodiversity (Walther, 2010). In conjunction with studies monitoring changes in taxonomic diversity, the monitoring of functional changes in species assemblages is crucial for understanding how natural systems are being impacted by habitat transformation (Fichaux et al., 2019). Changes in the composition of faunal assemblages that occur due to land-use change are dependent on the traits of the species (Wiescher et al., 2012). These traits are often related to the morphology, physiology, and ecology of the species, which influences its performance and functioning within the environment. With land-use intensification, environmental filtering may lead to the exclusion of species with traits

poorly adapted to a specific environment and support communities with more similar traits, leading to decreased functional diversity (Rigal et al., 2018).

Species with specialized diets, limited dispersal capacity and low fecundity have been identified as being most at risk of extinction due to landscape transformation (Gámez-Virués et al., 2015). In turn, this loss of functionally unique organisms may have detrimental effects on the functioning of the ecosystem (Butterfield and Suding, 2013). For example, decreased crop yields have been observed where functional homogenization occurred for flower visiting assemblages (Fründ et al., 2013). Conversely, diverse pollinator communities with species that vary in their flower height preference, time of visitation, and body size, lead to greater overall pollination success, highlighting the importance of functional complementarity (Hoehn et al., 2008). Thus, understanding the response of functional diversity to environmental change could

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further enable us to predict the consequences of land-use change on ecosystem functioning and ecosystem service provisioning (Winck et al., 2017). However, studies on the change in functional diversity across ecosystems are still scarce, especially for soil fauna (Salmon et al., 2014). Soil fauna plays a critical role within natural and agroecosystems (Marsden et al., 2020). Their role in organic matter decomposition, nutrient cycling, and the maintenance of soil structure (Menta and Remelli, 2020) is inherently linked to the health of the soil (Kibblewhite et al., 2008). As the taxonomic diversity of these taxa can be sensitive to changes in soil properties and micro-habitat characteristics (Fichaux et al., 2019), functional diversity is also expected to be sensitive to these changes (Fontanilla et al., 2019).

Monocultures of large-scale timber compartments consist of structurally less complex and more uniformly aged trees, with sparse undergrowth and lower litter quality compared to natural environments (Martin and He, 2015). Hence, compared to natural vegetation, they generally have negative effects on local biodiversity (Brockhoff et al., 2008). In landscapes where production areas provide poor habitat for species and contrast strongly with natural vegetation, the conservation of remnant vegetation is crucial for supporting biodiversity (Lindenmayer, 2019). Following this principle, conservation corridors consisting of remnant natural vegetation, primarily grasslands and indigenous forests, are conserved among timber production areas in South Africa (Samways and Pryke, 2016).

Previous studies on aboveground arthropods have shown that the conservation corridors conserve species-rich arthropod assemblages, but arthropod diversity is low in the plantation compartments (Pryke and Samways, 2012). In contrast, soil arthropods in these ecosystems are highly responsive to variation in the environment (Eckert et al., 2022b) and their assemblage composition is highly segregated between the natural areas and plantation compartments (Yekwayo et al., 2017; Eckert et al., 2019, 2022b). Yet, surprisingly, timber plantations in this region support equally high soil arthropod alpha- and beta-diversity as natural habitats (Eckert et al., 2022a). The impact of plantations on functional composition of soil arthropods is unknown but would provide important insight into the broader impact of plantation forestry on soil biodiversity and possible impacts on soil ecosystem functioning.

We assess the influence of exotic plantations on the traits and functional diversity of soil arthropods within a landscape mosaic. We have selected two focal taxa, namely ants (Formicidae) and springtails (Collembola). Springtails are one of the most important, abundant taxa within the soil (Chang et al., 2017), contributing to soil ecosystem functioning through litter decomposition and soil microstructure formation (Hopkin, 1997). Ants, on the other hand, are important components of the soil ecosystem as they constitute a major component of the soil animal biomass (Graham et al., 2009) and are one of the most important soil ecosystem engineers (Jouquet et al., 2006).

Our first objective is to examine how the taxonomic and functional richness of selected soil arthropod taxa differ between natural (grassland and indigenous forest) and transformed (eucalypt plantations) biotopes. Our second objective is to determine whether there are significant shifts in traits between the natural biotopes and the transformed biotopes for soil arthropods. To support the interpretation of the results, we compared the environmental conditions among the different biotope types. Understanding the response of soil arthropods to transformation and how their traits relate to different conditions, can assist plantation managers in conservation and landscape planning, ensuring the continued presence and services these functionally important fauna provide within the soil ecosystem.

2. Materials and methods

2.1. Study area and design

This study was conducted in two environmentally different geographical areas, namely the Midlands and Zululand (from here on

referred to as regions) (Fig. 1), in KwaZulu-Natal, South Africa. KwaZulu-Natal is characterized by a subtropical climate with warm, wet summers and cool, dry winters (Fairbanks and Benn, 2000). These two regions differ in terms of their elevation (the Midlands occurring at higher elevations compared to Zululand), their topography (the complex topography of the Midlands contrasts to the flat landscapes of Zululand), and soil type (the Midlands being dominated by deep, humic soils, whereas Zululand is dominated by shallow, sandy soils) (Bazelet and Samways, 2012; Eckert et al., 2022b). The high contrast in environmental conditions between these two regions, allow for an assessment of a wide range of conditions. Study sites were selected across several plantation estates within each region (Fig. 1, Table S1), and were situated within three biotope types: indigenous forest, grassland, and eucalypt plantation blocks (*Eucalyptus* spp.). Only plantation compartments in a mature growth stage were selected, i.e., >6 years into the rotation. The naturally occurring grasslands are maintained through prescribed burning every 2–5 years and grazing by domestic livestock (SANBI, 2014). For each biotope type, 18 sites were selected to maximize on landscape variation, totaling to 108 sites (Fig. 1), with focus being on a high level of spatial replication rather than temporal replication. Sites occurred within the elevation range of 917–1496 m a.s.l. in the Midlands, and 26–131 m a.s.l. in Zululand, with sites within the same biotope being >250 m apart. Sampling was conducted in the summer months of these regions, January–February 2018 for Midlands, and November–December 2018 for Zululand, when arthropods are most active.

2.2. Arthropod sampling and identification

Using pitfall trapping, four 300 ml plastic cups (9.5 cm diameter and 8 cm deep) were placed in a 2 m² grid at each site. The rim of the trap was level with the soil surface and filled with 50 ml of 60% ethylene glycol (containing a small amount of detergent to break the surface tension). Traps were left in the field for seven days, after which they were collected. Secondly, arthropods were collected directly using an aspirator within a randomly placed 1 m² quadrat of each site. This was conducted for 10 min by two people. Arthropods were also extracted using the Tullgren-Berlese extraction technique from 1 L of collected soil from each site (10 × 100 ml subsamples randomly collected). The soil was placed on top of a 0.5 mm aperture size mesh and left for 7 days to allow soil to dry and arthropods to be extracted into the attached 75% ethanol jar below the funnel. Specimens collected from all sampling methods were sorted into morphospecies (i.e., using external morphological characteristics only, without the use of identification keys) (Oliver and Beattie, 1996) (from here on referred to as ‘species’) and counted. Springtail and ant specimens were identified by taxonomic experts (A. Ndaba – for ants, CJS – for springtails). Springtails were identified to family level and ants to genus level. These taxonomic levels were of sufficient resolution for obtaining certain traits from the literature that do not vary substantially within these taxonomic groupings (section 2.3. Functional traits).

2.3. Functional traits

We divided the traits for ants and springtails into three categories: 1) morphological traits, 2) feeding strategy, and 3) habitat position (i.e., microhabitat occurrence). For ants, traits used in the analysis (Table S2) were derived from the literature (eye size, habitat position and feeding strategy) (Table S3) or measured in the laboratory (body length, femur length and darkness of the integument). For springtails, body length was measured in the laboratory, and all other traits (eye size, furca development, feeding strategy and habitat position) (Table S4) were determined using previous work on springtails traits (see Salmon et al., 2014; Raymond-Léonard et al., 2019). This was verified by a taxonomic expert (CJS) and a family-level key to the Collembola of South Africa (<https://collembola.co.za/identification-keys/>). Body length (ants and



Fig. 1. The focal plantation estates in a) the Midlands and b) Zululand, KwaZulu-Natal, South Africa. Illustrated are the study sites within indigenous forests (stars), grasslands (diamonds) and eucalypt compartments (circles). Plantation compartments (grey), non-plantation areas (white) and estate names indicated.

springtails) and femur length (ants) were measured using a Leica MZ7s dissecting microscope (Leica Microsystems, Germany). Where possible, six individuals per species were measured, and an average body length (for both taxa), and femur length (for ants only) was estimated. For polymorphic ants, individuals of varying sizes were selected, and an average body length calculated for those species. The body length of each ant species was measured dorsally from the tip of the mandible to the tip of the gaster, whereas femur length was measured in anterior view as it is a good proxy for overall leg length (Queiroz et al., 2015), and only the hind femur was measured. Ant integument colour was examined by the same method proposed by Bishop et al. (2016), using HSV (hue, saturation, and value) estimates, extracted from photos of the specimens using the image editing software *paint.NET* (v.4.0.3). Hue describes the dominant wavelength, while saturation indicates the amount of hue present in the colour and the value sets the amount of light in the colour (Bishop et al., 2016). For this study, only lightness (v, or value, in HSV) is analyzed (Bishop et al., 2016), ranging from 0 to 100, with a low V indicating a dark-coloured specimen, and a high V

indicating a light-coloured specimen (pers. comm. T. Bishop). For springtails, body length measurements were taken either dorsally (species belonging to Poduromorpha) or laterally (species belonging to Entomobryomorpha and Symphyleona) from the tip of the mandible to the tip of the last abdominal segment. Appendages such as antennae and furca were excluded in measurements for accuracy (Hóðar, 1996). Justification of traits included for this study (Table S5) and references thereof is provided (Table S3).

2.4. Biotic and abiotic environmental variables

To record other biotic and abiotic variables at each site, a 1 m² quadrat was randomly placed three times. These variables included vegetation height (m), vegetation cover (%), number of plant species, bare ground cover (%), leaf litter cover (%) and leaf litter depth (cm) and compositional characteristics (percentage of herbaceous, shrub and grass cover). The soil compaction, moisture and pH of each site was recorded using a soil penetrometer (Lang Penetrometer, Inc.) and a soil

moisture and pH meter (Kelway, Inc.) within the quadrat. In a 5 m radius surrounding the quadrat (within 80 m²), the amount of shade (%), deadwood cover (%) and rock cover (%) was estimated and recorded.

2.5. Data analyses

Data analyses were performed separately for springtails and ants of each region, using pooled data from all sampling methods. All data analysis was conducted using R (version 4.2.2) (R Core Team, 2016).

Species accumulation curves were created for each arthropod taxon, separately per biotope type and region, using the *vegan* package (Oksanen et al., 2013) (Fig. S1) to determine whether sampling was sufficient (Gotelli and Colwell, 2001). Species richness estimators (Table S7) were calculated using the *fossil* package (Vavrek, 2012). Chao2 and Jackknife2 were selected as species richness estimators as they are suitable for estimating species diversity of highly diverse communities (Colwell and Coddington, 1994).

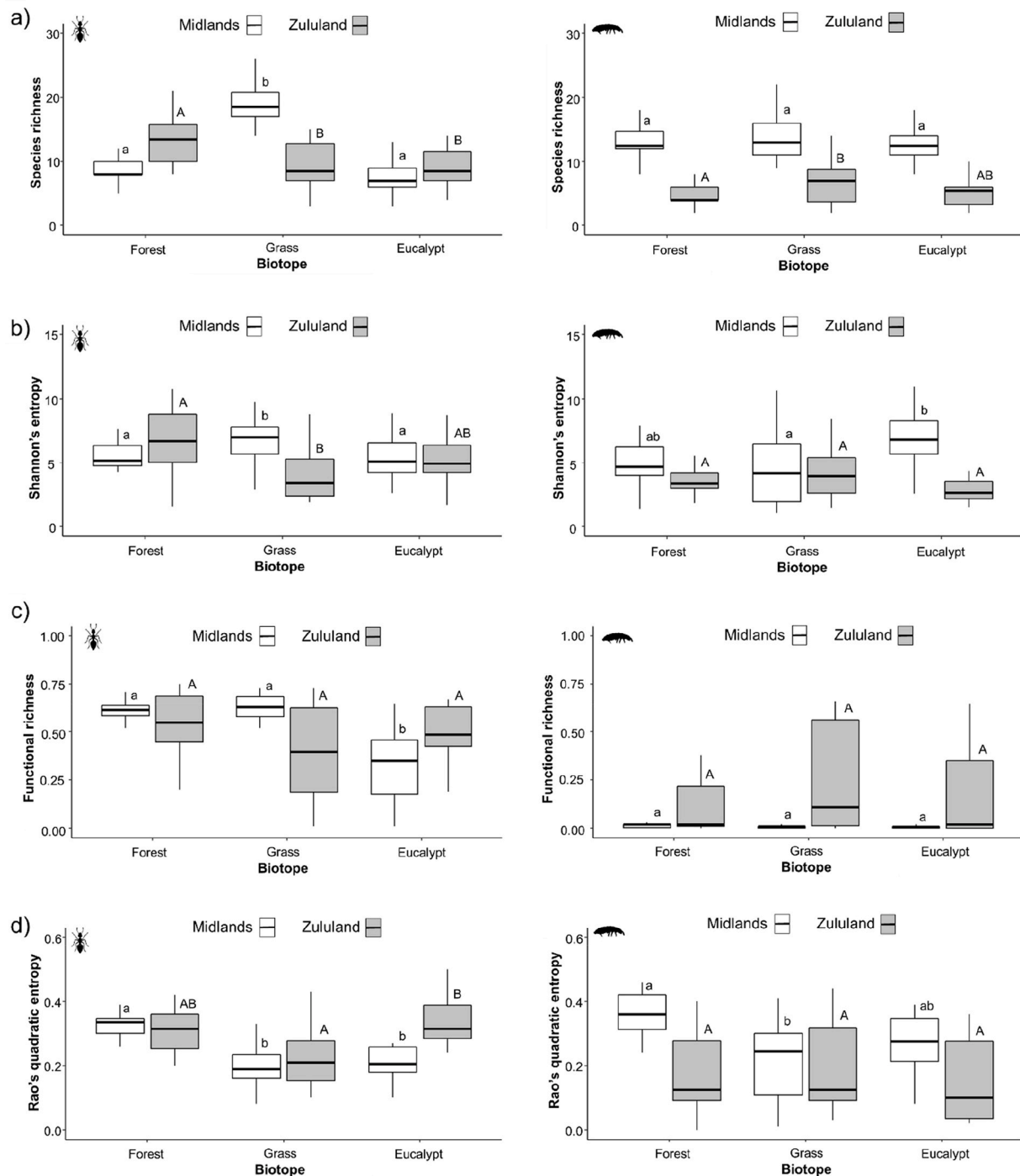


Fig. 2. Boxplots with central horizontal line indicating the median value, the boxes indicating the interquartile range, and the whiskers indicating standard deviation within each biotope type. Illustrated is the a) species richness, b) Shannon's entropy (taxonomic evenness), c) functional richness and d) Rao's quadratic entropy (functional diversity) of ants and springtails in the Midlands and Zululand. Means with small letters for Midlands and capital letters for Zululand in common between indigenous forests (Forest), grasslands (Grass) and eucalypt compartments (Eucalypt), are not significantly different at $P < 0.05$.

2.5.1. Comparison of taxonomic and functional richness in plantation compartments and natural areas

Local species richness was estimated for each site using morpho-species richness (from here on referred to as ‘species richness’). Functional richness (representing the dispersion of species in the functional space) was calculated in R using the *dbFD* function of the *FD* package (Laliberté and Shipley, 2011), whereas Shannon’s entropy (representing taxonomic evenness) and Rao’s quadratic entropy (representing alpha-functional diversity) was calculated using *hillR* package (Li, 2018). Traits used for the calculation of functional richness and Rao’s quadratic entropy for ants (Table S2) and springtails (Table S4) are provided. For species richness, Shannon’s entropy, functional richness and Rao’s quadratic entropy, data normality distribution was assessed using Shapiro-Wilk tests in the *nortest* package (Gross and Gross, 2009), and family distribution was assessed with the *car* and *MASS* packages using quantile-quantile plots (Ripley et al., 2013; Fox and Weisberg, 2017), and spatial autocorrelation was assessed based on Mantel tests using the *ade4* package (Dray and Dufour, 2007).

Generalized linear mixed models (GLMMs) were conducted using the *lme4* package (Bates et al., 2014). Biotope type was included as a fixed parameter in all models and plantation estate as a random parameter to account for spatial autocorrelation (Fig. 2). Pairwise Wilcoxon tests for non-parametric data using the *pairwise.wilcox.test* function with Bonferroni correction were conducted where significant effects occurred. Boxplots were constructed using the *ggpubr* package (Kassambara, 2017).

2.5.2. Trait associations with biotopes

To determine the relationship between the traits of species and biotope type, a fourth-corner analysis (Ter Braak et al., 2012) was conducted using a trait generalized linear model with the *mvabund* package (Wang et al., 2020). These models were constructed using three data sets: an abundance matrix, biotope type matrix, and the trait matrix of the assemblage. The abundance matrix best fitted a negative binomial distribution. To reduce the number of uninformative associations, all models included the LASSO penalty to reduce the trait-biotope association coefficients to 0 when the effects were small (Wang et al., 2020). The *lattice* package (Sarkar and Sarkar, 2007) was used to visually show the relationship and its strength between individual traits and biotope type. Fourth-corner analysis was conducted separately for the following categories: morphological traits, feeding strategy and habitat position (Tables S2 and S4). Analyses were conducted separately for springtails and ants for each region.

2.5.3. Environmental differences between biotope types

To determine differences of environmental variables (section 2.4) between biotope types of each region, linear models (LM’s) were performed. For pairwise comparisons, Tukey post-hoc tests were performed where significant effects occurred, using the *multcomp* package (Bretz et al., 2002, 2008). Results of pairwise difference are shown in Table S6 for the Midlands and Zululand.

3. Results

Midlands samples consisted of 15 678 ant individuals, in 87 morphospecies in 19 genera in six sub-families. Zululand samples consisted of 10 053 ant individuals, consisting of 74 morphospecies in 25 genera in 6 sub-families. For springtails, Midlands samples consisted of 63 719 individuals, in 87 morphospecies in 11 families and three orders. Zululand’s samples consisted of 5119 springtail individuals, in 47 morphospecies in 11 families and 3 orders.

3.1. Taxonomic and functional richness in the different biotopes

3.1.1. Formicidae

Ant species richness (Fig. 2A) and taxonomic evenness (Fig. 2B)

differed significantly between biotopes in the Midlands (Table 1), with grasslands having significantly more ant species and higher taxonomic evenness compared to both the indigenous forests and eucalypt compartments. Ant species richness in Zululand was significantly higher in indigenous forests compared to both the grasslands and eucalypt compartments (Fig. 2A, Table 1). In Zululand, taxonomic evenness in indigenous forests was significantly higher than in grasslands, but taxonomic evenness in eucalypt compartments did not differ from that in either indigenous forests or grasslands (Fig. 2B, Table 1). Ant functional richness in the Midlands was significantly lower in eucalypt compartments than in both indigenous forests and grasslands (Fig. 2C, Table 1). In Zululand, no significant difference was found for ant functional richness between biotopes (Fig. 2C, Table 1). Ant functional diversity in the Midlands was significantly higher within the indigenous forests compared to both the grasslands and eucalypt compartments (Fig. 2D, Table 1). In Zululand, ant functional diversity was significantly higher in eucalypt compartments than in grasslands, while indigenous forests did not differ from either the grasslands or eucalypt compartments (Fig. 2D, Table 1). Summary of results shown in Table 2.

3.1.2. Collembola

Springtail species richness did not differ significantly between biotopes in the Midlands (Fig. 2A, Table 1). In Zululand, springtail species richness was significantly higher in the grasslands compared to the indigenous forests, while the eucalypt compartments did not differ from either the indigenous forests or grasslands (Fig. 2A, Table 1). Springtail taxonomic evenness did not differ between biotopes in the Midlands or in Zululand (Fig. 2B, Table 1). Springtail functional richness did not differ significantly between biotopes in the Midlands and Zululand (Fig. 2C, Table 1). Springtail functional diversity in the Midlands was significantly higher within the indigenous forests compared to the grasslands (Fig. 2D, Table 1). Additionally, springtail functional diversity was significantly higher within the indigenous forests compared to both the grasslands and eucalypt compartments, with the grasslands and eucalypt compartments showing no significant difference (Fig. 2D, Table 1). In Zululand, no significant difference was found for springtail functional diversity between biotopes (Fig. 2D, Table 1). Summary of results shown in Table 2.

3.2. Trait-association of formicidae and collembola to biotope type

3.2.1. Formicidae

Generally, specific taxonomic and trait associations with the biotopes varied between the two regions. However, there were clear shifts in trait composition between biotopes in both regions. In the Midlands, darker ants with reduced eyes were associated with the indigenous forests, while ants with smaller bodies were associated with the eucalypt

Table 1
Effect of biotope on species richness, Shannon’s entropy, functional richness and Rao’s quadratic entropy for ants (Formicidae) and springtails (Collembola) in the Midlands and Zululand. Significant effects at $p < 0.05$.

Response	Midlands		Zululand	
	χ^2	df	χ^2	df
Formicidae				
Species richness	72.41***	2	16.69***	2
Shannon’s entropy	10.10**	2	11.23**	2
Functional richness	10.80**	2	2.18	2
Rao’s quadratic entropy	26.08***	2	7.77*	2
Collembola				
Species richness	1.38	2	6.70*	2
Shannon’s entropy	4.71	2	6.49	2
Functional richness	0.43	2	0.18	2
Rao’s quadratic entropy	6.69*	2	0.83	2

Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 2

Summary of results obtained during research study. For univariate results, letters for the Midlands (small) and Zululand (capital) show the pairwise comparisons for each response variable between biotopes within each region, per taxon. Biotopes indicated as indigenous forests (For), grasslands (Grass) and eucalypt compartments (Euc). Letters in common are not significantly different at $P < 0.05$. For multivariate responses, trait associations are shown as a positive (+), negative effect (−), no association (No), or not applicable (n/a).

Response variables	Formicidae						Collembola					
	Midlands			Zululand			Midlands			Zululand		
	For	Grass	Euc	For	Grass	Euc	For	Grass	Euc	For	Grass	Euc
UNIVARIATE												
Species richness	a	b	a	A	B	B	a	a	a	A	B	AB
Shannon's entropy	a	b	a	A	B	AB	ab	a	b	A	A	A
Functional richness	a	a	b	A	A	A	a	a	a	A	A	A
Rao's quadratic entropy	a	b	b	AB	A	B	a	b	ab	A	A	A
MULTIVARIATE												
Morphological traits												
Moderate/large eye size	−	No	No	No	−	No	−	+	No	No	No	−
Absent/small eye size	+	No	No	No	+	No	+	−	No	No	−	+
Colour (lightness)	−	−	No	−	+	No	n/a	n/a	n/a	n/a	n/a	n/a
Femur length	No	No	No	No	+	−	n/a	n/a	n/a	n/a	n/a	n/a
Body length	−	+	−	−	−	+	+	No	−	No	−	+
Moderate/large furca	n/a	n/a	n/a	n/a	n/a	n/a	No	No	−	−	No	No
Absent/small furca	n/a	n/a	n/a	n/a	n/a	n/a	−	No	+	+	No	No
Feeding strategy												
Saprovore	−	+	No	−	+	No	n/a	n/a	n/a	n/a	n/a	n/a
Predator	+	−	No	−	−	+	n/a	n/a	n/a	n/a	n/a	n/a
Generalist	No	No	No	No	No	No	n/a	n/a	n/a	n/a	n/a	n/a
Sucking	n/a	n/a	n/a	n/a	n/a	n/a	No	No	+	+	No	−
Chewing	n/a	n/a	n/a	n/a	n/a	n/a	No	No	−	−	No	+
Habitat position												
Vegetation-dwelling	n/a	n/a	n/a	No	No	No	No	No	No	No	No	No
Litter-dwelling	No	−	+	No	No	No	−	No	+	+	No	−
Soil-dwelling	No	+	No	−	+	No	+	−	No	No	−	+

compartments and ants with larger bodies were associated with the grasslands (Fig. 3A). In Zululand, darker ants were associated with the indigenous forests, while lighter ants were associated with the grasslands (Fig. 3A). Additionally, ants with short and long legs were associated with the eucalypt compartments and grasslands, respectively, in Zululand (Fig. 3A). Ants with large and small bodies were also associated with the eucalypt compartments and grasslands, respectively (Fig. 3A). In the Midlands, predatory ants were associated with the indigenous forests and saprovores with the grasslands, while no association was found for the eucalypt compartments (Fig. 3B). In Zululand, predatory ants were associated with the eucalypt compartments and saprovores with the grasslands, while saprovores were negatively associated with the indigenous forests (Fig. 3B). In the Midlands, litter-dwelling ants were associated with the eucalypt compartments, and soil-dwelling ants to the grasslands, while no association was found for the indigenous forests (Fig. 3C). In Zululand, soil-dwelling ants were associated to the grasslands but negatively associated to the indigenous forests, while no association was found for the eucalypt compartments (Fig. 3C). Summary of results shown in Table 2.

3.2.2. Collembola

In the Midlands, springtails with moderate/large eyes were associated with the grasslands (Fig. 4A). Springtails with absent/small eyes and large bodies were associated with the indigenous forests, while springtails with absent/small furcas and small bodies were associated with the eucalypt compartments (Fig. 4A). In Zululand, small-bodied springtails were associated with the grasslands, while springtails with absent/small furcas were associated with the indigenous forests (Fig. 4A). Additionally, springtails with absent/small eyes and large bodies were associated with the eucalypt compartments in Zululand (Fig. 4A). In the Midlands, springtails with sucking mouth parts were associated with the eucalypt compartments, while no associations were

found for the indigenous forests or grasslands (Fig. 4B). In Zululand, springtails with sucking mouth parts were associated with the indigenous forests and springtails with chewing mouth parts were associated with the eucalypt compartments (Fig. 4B). No association between mouth parts and the grasslands were found in Zululand (Fig. 4B). In the Midlands, soil-dwelling springtails were positively associated to the indigenous forests, but negatively associated with the grasslands (Fig. 4C). Additionally, litter-dwelling springtails were associated with the eucalypt compartments in the Midlands (Fig. 4C). In Zululand, soil-dwelling springtails were positively associated to the eucalypt compartments, but negatively associated with the grasslands (Fig. 4C). Additionally, litter-dwelling springtails were associated with the indigenous forests in Zululand (Fig. 4C). Summary of results shown in Table 2.

3.3. Environmental differences between biotope types

The three biotopes differed in their biotic and abiotic environmental variables in the Midlands and Zululand (Table 3). In the Midlands, bare ground cover was significantly higher within the grasslands, followed by the eucalypt compartments and indigenous forests (Table S6). Dead wood cover and leaf litter cover was significantly higher within the eucalypt compartments, followed by the indigenous forests and grasslands (Table S6). Plant species richness was significantly higher within the indigenous forests, followed by the grasslands and eucalypt compartments (Table S6). Soil compaction was significantly higher within the indigenous forests, followed by the eucalypt compartments and grasslands (Table S6). Soil moisture was significantly higher within the eucalypt compartments, followed by the grasslands and indigenous forests (Table S6). Soil pH was significantly higher within the indigenous forests and grasslands compared to the eucalypt compartments (Table S6). Vegetation cover was significantly higher within the

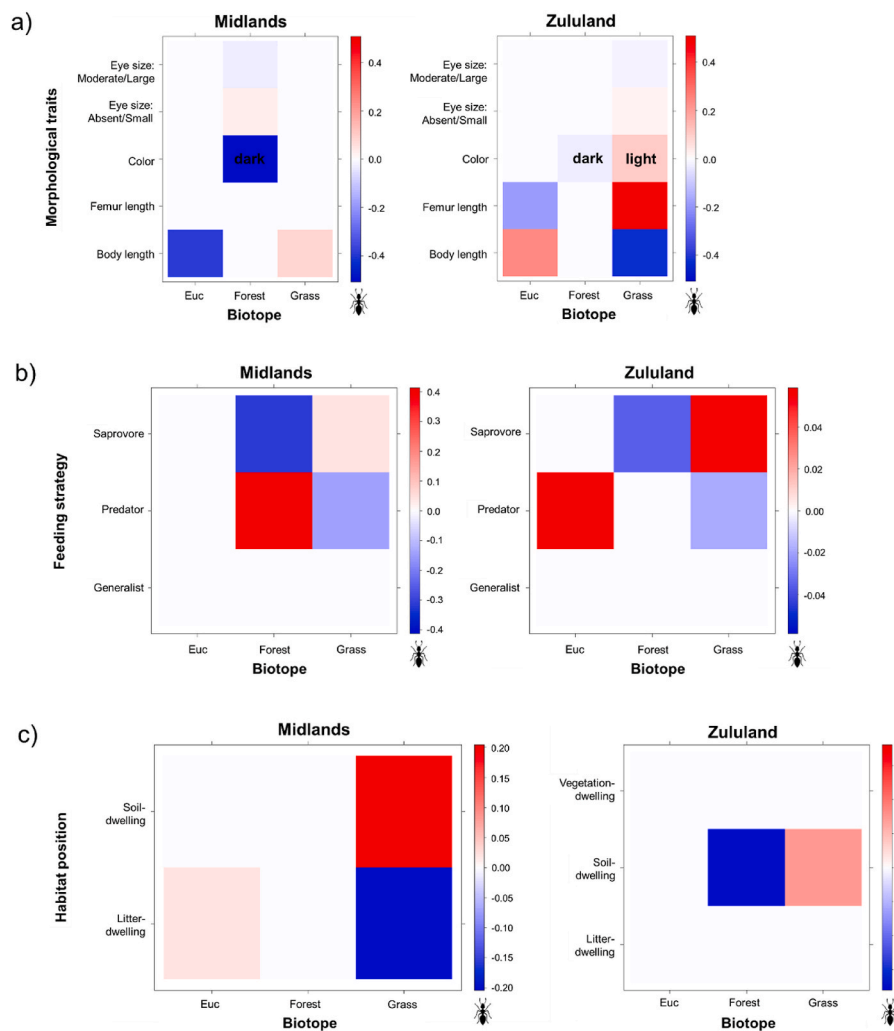


Fig. 3. Fourth-corner analysis results of a) morphological traits, b) feeding strategy and c) habitat position of ants in the Midlands and Zululand between indigenous forests (For), grasslands (Grass) and eucalypt compartments (Euc). Red indicates a positive association and blue indicates a negative association, with the colour intensity showing the strength of the relationship.

grasslands, followed by the indigenous forests and eucalypt compartments (Table S6).

In Zululand, bare ground cover was significantly lower within the indigenous forests and eucalypt compartments compared to the grasslands (Table S6). Dead wood cover was significantly higher within the indigenous forests and eucalypt compartments compared to the grasslands (Table S6). Leaf litter cover was significantly higher within the eucalypt compartments, followed by the indigenous forests and grasslands (Table S6). Soil compaction was significantly lower within the indigenous forests and grasslands compared to the eucalypt compartments (Table S6). Soil moisture was significantly lower within the grasslands and eucalypt compartments compared to the indigenous forests (Table S6). No significant difference in soil pH was found between the three habitat types (Table S6). Vegetation cover was significantly higher within the grasslands, followed by the indigenous forests and eucalypt compartments (Table S6).

4. Discussion

4.1. Taxonomic and functional alpha-diversity in contrasting biotopes

Taxonomic and functional alpha-diversity is generally higher within more heterogeneous, diverse systems (Winck et al., 2017), as they provide greater niche opportunities for functionally divergent species.

Additionally, landscape transformation for agriculture drives biotic and functional homogenization (Gámez-Virués et al., 2015). Similar patterns have been observed for timber plantations (Martello et al., 2018), yet in our study region, equally high levels of alpha- and beta-diversity have been recorded in plantations (*Eucalyptus* and *Pinus*) compared to natural habitats (indigenous forest and grassland) for ants and springtails (Eckert et al., 2022a). Here, we expand on these results by showing that exotic *Eucalyptus* plantations can also harbour similar levels of functional diversity compared to either indigenous forests or grasslands (or in some cases both) for ants and springtails, even though it is a transformed and relatively homogenous system. Our results agree with Winck et al. (2017), showing eucalypts can harbour similar levels of functional diversity compared to indigenous forests or grasslands, although assemblage compositions can still differ. As the stability of an ecosystem is driven by the co-existence of soil organisms performing different functions to maintain equilibrium (Griffiths et al., 1997; Kostin et al., 2021), our results indicate potentially good ecosystem stability and functioning in the plantations. This raises the question whether there are functional traits that enable species to persist within these contrasting biotopes.

4.2. Functional trait adaptations in contrasting biotopes

Species persistence and fitness in each environment is regulated by

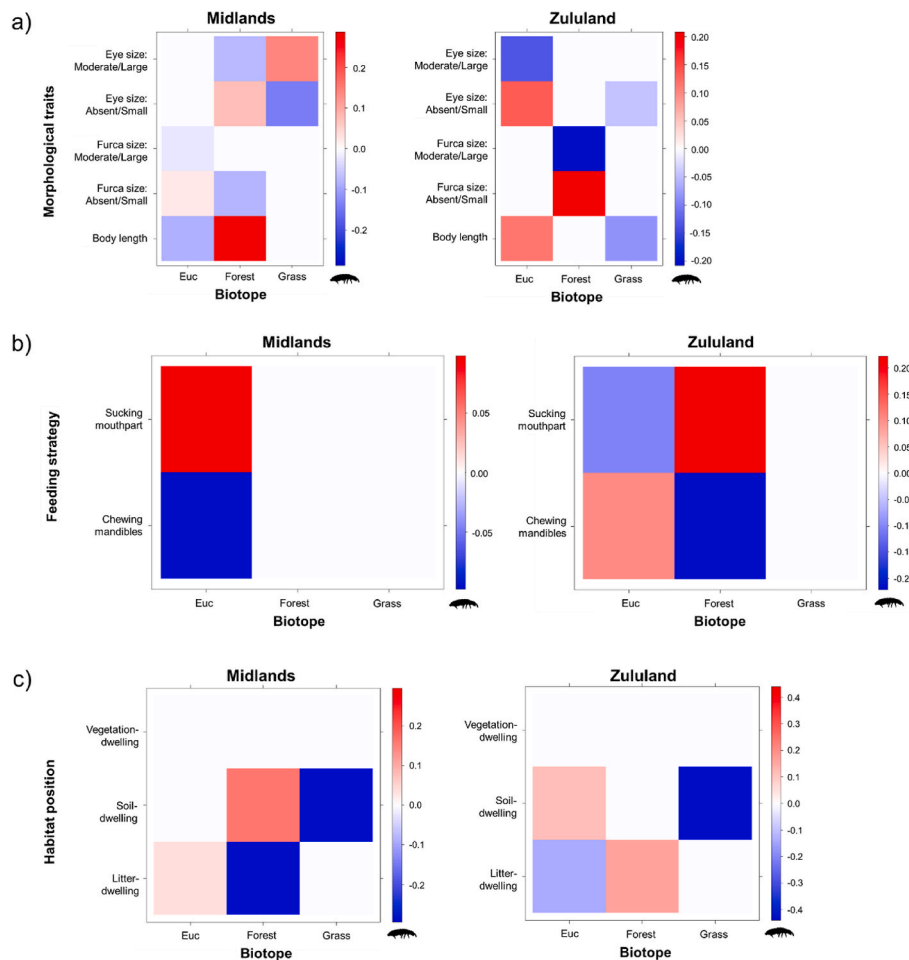


Fig. 4. Fourth-corner analysis results of a) morphological traits, b) feeding strategy and c) habitat position of springtails in the Midlands and Zululand between indigenous forests (For), grasslands (Grass) and eucalypt compartments (Euc). Red indicates a positive association and blue indicates a negative association, with the colour intensity showing the strength of the relationship.

Table 3

Effect of biotope type on biotic and abiotic environmental variables across habitat types within each region using linear models (LM's). Reported is the F-value, with significant effects at $p < 0.05$. Pairwise comparisons reported in Table S6.

Environmental variable	Midlands	Zululand
Bare ground cover (%)	23.53***	21.02***
Dead wood cover (%)	222.01***	38.75***
Grass cover (%)	106.02***	311.04***
Herbaceous cover (%)	39.25***	29.40***
Leaf litter cover (%)	242.26***	45.07***
Leaf litter depth (cm)	199.18***	37.49***
Number of plant species	94.69***	24.54***
Rock cover (%)	6.24**	n/a
Shade cover (%)	273.88***	333.97***
Shrub cover (%)	26.36***	4.31*
Soil compaction (psi)	23.53***	33.47***
Soil moisture (%)	1.92	13.27***
Soil pH	10.93***	1.67
Tree density (%)	108.31***	93.64***
Vegetation cover (%)	197.37***	28.78***
Vegetation height (cm)	47.52***	18.58***

Significance codes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

the compatibility of the species' traits to the environmental conditions therein (Wiescher et al., 2012). The associations that we observed between arthropod traits and the three biotopes were not always clear, and in many cases the patterns were not consistent between the two regions.

This may be due to differences between the regions in climate (sub-tropical in Zululand and temperate in the Midlands), topographic complexity (low complexity in Zululand and high in the Midlands) and soil type (shallow, nutrient poor sandy soils in Zululand and deep, nutrient rich soils in the Midlands). The collective differences in the above-mentioned factors may contribute to environmental and micro-climate variances that favour a completely different set of traits within similar biotopes, but in different regions. Another reason for the lack of clear associations may be that certain individual traits are related to multiple life history aspects, e.g., body size relates to dispersal ability, habitat preference and physiological/ecological tolerances (Bishop et al., 2016), which could obscure single trait-environment relationships. Nonetheless, in each region, we saw clear shifts in the prevalence of traits between the different biotopes.

4.2.1. Formicidae

A common observation in ant trait-based studies, is that in cooler, closed-canopy environments, individuals are darker in colour, whereas the opposite is observed in warmer, open environments (Gibb et al., 2015). Colour in ants is related to thermoregulation, as darker coloured organisms heat up faster than lighter coloured organisms (Bishop et al., 2016). Our results are similar to these findings, as ants in the indigenous forests were darker than in grasslands and eucalypt compartments in both regions. We also found soil-dwelling species to be associated with the grasslands in both regions. As these grasslands undergo frequent disturbance in the form of burning and grazing (Bazelet and Samways,

2012) and have lower shelter opportunities from leaf litter cover and shade than the wooded biotopes, it may be an adaptive advantage to be soil-living to avoid extreme conditions. Ant body size is also partly related to thermoregulation, as larger bodies enable organisms to lose heat slower than smaller bodied organisms, and larger organisms would be expected in cooler, canopied environments. This was observed in Zululand, with smaller individuals in grassland compared to the wooded biotopes. However, the opposite trend was seen in the Midlands. Body size, however, also relates to manoeuvrability in the environment, with smaller bodies being more beneficial in more complex environments (Farji-Brener et al., 2004). This may have explained the prevalence of large-bodied ants in the grassland. Although it is difficult to say with certainty how arthropods perceive the complexity of the studied vegetation types, the grassland generally had greater bare ground cover, less leaf litter cover, and less dead wood cover than the wooded biotopes in both regions (Table S6), suggesting that the litter and topsoil layer in grassland may be less structurally complex for the focal organisms. Regarding femur length, several studies suggest ground cover complexity is correlated to femur lengths (Wiescher et al., 2012), with shorter femur lengths being more beneficial in more complex environments. Long legs were observed in the grasslands and short legs in eucalypt compartments in Zululand, corresponding with the expected pattern if the grassland soil surface is perceived as structurally simpler than in plantations. We observed no association between femur length and habitat type in the Midlands.

As grasslands have no tree canopy cover, limited litter cover and experience burning and grazing as part of the management regime (Bazelet and Samways, 2012), generalist foraging ants would be better adapted for these conditions than ants with more specialized diets. Species with unspecialized diets are better adapted to dealing with the dynamic nature of the grasslands and have a wider range of habitat tolerance and are less sensitive to disturbances than species with specialized diets (Andersen, 1995). However, we found saprovores dominated in both regions within the grasslands, the reason for which is unclear. Predatory ants were associated with the indigenous forests and eucalypt compartments in the Midlands and Zululand, respectively. Complex environments can provide a greater variety of resources to exploit (Pacheco and Vasconcelos, 2012), with lower inter-annual disturbance. Indeed, disturbance levels are low in both forest and eucalypt plantations, as forests are virtually undisturbed, and plantations undergo disturbance only once per 8–10-year rotation. This may allow predatory ant populations to build up.

4.2.2. *Collembola*

Litter-dwelling and vegetation-dwelling species are well adapted to environmental conditions aboveground, and to conditions in open environments, as the microclimatic conditions can be more variable and often drier than belowground or in sheltered, wooded environments (Makkonen et al., 2011). Therefore, these species are more drought-tolerant than belowground species (Makkonen et al., 2011). We observed this pattern in this study, as in both regions, springtails associated with subterranean life were negatively associated with the grasslands, whereas they were positively associated with the indigenous forests and eucalypt compartments, respectively in the Midlands and Zululand. Body size is also linked with protection against desiccation and frost (Bokhorst et al., 2012), with larger bodies being better adapted to above-ground living. However, we did not observe these body size associations, with larger-bodied springtails being associated with indigenous forests in the Midlands and to eucalypt plantations in Zululand. Additionally, springtails in structurally simpler, more open environments, such as grasslands, generally have well-developed eyes and furcas (Salmon et al., 2014), aiding in predator detection and avoidance, whereas in more complex, closed environments, such as woodlands, they have absent or reduced eyes and furcas (Salmon et al., 2014). This is because a complex ground cover, like a dense litter layer, offers protection against predation (Hossie and Murray, 2010), making predator

avoidance relatively less important. The eye size associations from this study agreed with this prediction, as the wooded biotopes (eucalypt plantations in Zululand and indigenous forest in the Midlands) had a prevalence of individuals with small or absent eyes, and grassland was generally associated with moderate or large eyes. For furca development, we found that springtails with absent/reduced furcas were associated to the indigenous forests in Zululand, and associated to the eucalypt compartments in the Midlands, again conforming to the idea that the deep litter layer in these biotopes play an important role in selecting for certain species traits in the different habitats.

Chewing/grinding mandibles are associated with a generalist diet (Hopkin, 1997) and can consist of harder items such as plant tissue and coarse leaf litter (Malcicka et al., 2017), whereas suctorial/piercing mouthparts are associated with feeding on liquids (Hopkin, 1997) from micro-organisms (Hoskins et al., 2015). We found springtails with suctorial/piercing mouthparts to be positively associated with the *Eucalyptus* plantations in the Midlands, and positively associated with the indigenous forests in Zululand. As the leaf litter layer likely offers a great variety of food resources, such as fungi and mould (Salmon and Ponge, 2001), this suggests that the dense litter layer in the wooded biotopes may indirectly play a role determining different mouthpart. In contrast, springtails with chewing mandibles were negatively and positively associated with the eucalypt compartments in the Midlands and Zululand, respectively. Here, other factors, such as soil chemical composition and moisture may also have played a role in providing the optimal conditions for springtail food availability and quality (Erlandson et al., 2016).

5. Conclusion

We demonstrate that, although plantations are structurally and compositionally less complex than natural systems, *Eucalyptus* plantations in this region support equally high taxonomic and functional diversity as the indigenous forests and grasslands. It appears that the high functional diversity in all biotopes is partly due to species in the contrasting biotopes possessing traits that predispose them to persist and thrive in the different conditions. This has important implications for ecosystem stability and functioning the *Eucalyptus* plantations, given the presence of a functionally rich component of soil biodiversity within the plantations. This signifies that processes such as litter breakdown and nutrient cycling may still be well-maintained in these production areas, although this would need to be examined in future work. Furthermore, we show that remnant natural vegetation types interspersed with the plantations are beneficial for maintaining historic soil arthropod taxonomic and functional diversity at the landscape scale. Therefore, although the various landscape elements are environmentally different, together they can be seen as an agroecological land mosaic, consisting of various environmental conditions suitable for different sets of soil arthropods, which contribute to overall landscape biodiversity. Promoting landscape and environmental heterogeneity within a production landscape can thus provide a wider range of niches for arthropods with different habitat requirements to exploit, which in turn, can increase functional diversity across the production landscape.

Author contributions

ME, RG, JSP and MJS conceived the ideas and designed the methodology. Data collection was performed by ME. CJS identified Collembolan specimens. ME, RG, JSP lead the analyses on the data. The first draft of the manuscript was written by ME. All authors discussed the results and contributed to the final manuscript.

Ethical approval

Ethical approval was not required for this study, as it involves sampling arthropods, which do not fall under the classification of higher

invertebrates according to SANS 10386:2008.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.soilbio.2023.109180>.

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